Resume
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The year 2023 marks a turning point for researchers, where a discovery mindset and motivation for innovation are prioritized over technical skills more than ever before due to the advancement of Generative AI. I have been following the development in the GenAI field very closely. My research will use AI in every step.

Research/Work

- **Research Assistant for Precision Swine Production - University of Missouri 07/2024 - 12/2024**
- Developed solutions to predicate estrous time in swine: configured MCU, and LIDAR/IR cameras with codes to become an inspection robot; trained in models for imaging segmentation and transformer neural network. Filled an NSF fellowship proposal, a two-page extended abstract proposed the ground true uncertainty issue might be addressed by using microneedle patch for continuous estradiol monitoring
- **Bioinformatics Research Assistant - Florida Atlantic University/Texas Tech University, 03/2023 - 06/2023, 04/2024 - 06/2024**
- Performed mutation and transposable element between samples from whole exome/genome data,, using GATK/BWA/Samtools/Bcftools/Pysam/Snpeff/customized python code, project-based short term contractor
- **Research Data Assistant - IU School of Medicine, 06/2021 - 07/2022**
- Analyzed over 50 various RNA-seq projects at sequencing core, included: processing raw RNA sequencing data, mapping, sample outlier detection, batch effect removing, various DGE analysis, pathway analysis, splicing analysis, troubleshooting
- Generated html report with the newest visualization tools/plots for the users
- Processed raw WES data, SNP, INDEL analysis
- Developed a GUI database
- Provided consultation to researchers regarding the analysis result
- Mentored undergraduate and master students

Education

- **Arizona State University, 01/2025 – present, M.S. Electrical Engineering, online**
  Took VLSI design, Digital Signal Processing (DSP) courses

- **San Diego State University, 01/2025 – present, M.S. Medical Physics, onsite**
  Courses in medical physics, shadowed in a MEMS lab with project using electrochemistry and graphene to monitor glucose level in real time

- **University of Missouri Columbia, 08/2024-12/2024, CS master student**
  Took courses in VHDL digital logic devices, CMOS, Virtuoso, Vivado
- **Purdue University, 08/2023-05/2024, non-degree student taking medical physics courses,**
  Took courses about how imaging begging generated through X-ray, CT, PET, MR, or ultrasound; conducted MR research that isolated non-water proton signals, led in a poster presentation at the school conference; developed TG51 online calculator for radiation dosage calculation led to oral presenting in *regional* conference

2017-2021

- **Indiana University – Purdue University Indianapolis (IUPUI), B.S. in Biomedical Informatics**
  150 credits in life science, statistics, computer science, bioinformatics, Internship to identify potential pathways and markers related to mitochondria in RGC cells from single cell sequencing data.
- **Saint Louis Community College, University of Nebraska Omaha, Ivy Tech Community College**
  NSF funded 16 credits hands-on training in biotechnology, Indiana General Education Transfer Core Certificate

Projects

- **Genomics Bioinformatics (2020-2023): Whole genome SNP/INDEL calling, transposable-element mapping, large scale RNA/miRNA/miRNA/lncRNA pipelines, multi-generation family inheritance mutation calling, possible biological meaning with literature search; single cell functional check** https://ax3301.github.io/RGC/rgc_2021.html**; dashboard development** https://ax3301.github.io/PDAC_2022
- **Web Scraping + Functional call: Toolkit for new medical physics students** https://ax3301.github.io/Medical_Physics_Toolkit_23/
- **NSF GRFP Proposal 2023: Large language models for advancements in biomedical research** https://ax3301.github.io/RGC/NSF_2023.pdf
- **NSF GRFP Proposal 2024: Precision livestock farming using engineering strategies in swine Barns** https://ax3301.github.io/RGC/NSF_2024.pdf
- **Water signal suppression in NMR imaging** https://github.com/ax3301/RGC/blob/main/NMR_2024.ipynb
- **Swine Vision: Imaging segmentation and time serious transformer neural network in swine** https://github.com/ax3301/Pig_Vision_2024
- **NSF New Grants Tracker (Developed by AI Agent)** https://nsf-grant-tracker-2025.sequencingmaster.com/

Publication, Poster, Oral Presentation

- Chang H. (2021). A comprehensive list of undergraduate bioinformatics programs in the United States. https://doi.org/10.31219/osf.io/9g5wu
- Chang H., Q H. (Mar 2024). Visualization and algorithm for in vivo piSSelMQC signal processing in chemical shift imaging experiments: Recovering multi-quantum coherence transfer pathways. Presented at 2024 HSCI Purdue University Annual Research Retreat https://ax3301.github.io/RGC/HSCI_032024.pdf
- Chang H. (Apr 2024). The purpose of developing an online TG51 calculator for quick beam measurement: To provide a convenient and accessible tool for trainees. Oral presentation at: American Association of Physicists in Medicine ORVC 2024 Spring Meeting. https://ax3301.github.io/RGC/ORVC_2024.pdf

Technical Skills

Languages & Tools: Python, R, MATLAB, VHDL, SystemVerilog, HTM, Bash
Hardware & EDA: Microcontrollers, LiDAR/IR sensors, Virtuoso, Vivado
AI & ML: PyTorch, Transformers, OpenCV, Prompt Engineering
Bioinformatics: GATK, BWA, SAMtools, BCFtools, SnpEff, Pysam, Seurat, DESeq2
Data & Visualization: Pandas, Plotly, Dash, Matplotlib, ggplot2